4-2-26 (WEEK 6) (A little bit out of order due to tech issues — Geom dist on page 2 occurs before this part) Shun/翔海 (@shun4midx)

# NEGATIVE BINOMIAL DISTRIBUTION

Denoted as: $X \sim NB(r, p)$, $0 < p \leq r-1$, $r \in \mathbb{N}$

Setting: Repeat indep trials w/ success prob $p$
Count # trials until $r$ successes: $X \in \{r, r+1, r+2, \dots\}$

## OBSERVATION

$NB(1, p) = Geom(p)$

# NEGATIVE BINOMIAL DISTRIBUTION QUANTITIES

$\mathbb{P}(X=n) = \binom{n-1}{r-1} p^r (1-p)^{n-r}$ (r successes + n-r failures + choose r-1 successes in n-1 trials)

Expectation

$$\mathbb{E}[X] = \sum_{n=r}^{\infty} n\mathbb{P}(X=n) = \sum_{n=r}^{\infty} n\binom{n-1}{r-1} p^r (1-p)^{n-r} = \sum_{n=r}^{\infty} \frac{n(n-1)!}{(n-r)!(r-1)!} p^r (1-p)^{n-r} = \sum_{n=r}^{\infty} \frac{rn!}{r!(n-r)!} p^r (1-p)^{n-r} = \sum_{n=r}^{\infty} r\binom{n}{r} p^r (1-p)^{n-r} = \frac{r}{p} \sum_{n=r}^{\infty} \binom{n+1-1}{r+1-1} p^{r+1} (1-p)^{(n+1)-(r+1)}$$

$$= \frac{r}{p} \sum_{n=r+1}^{\infty} \binom{n-1}{r+1-1} p^{r+1} (1-p)^{n-(r+1)}$$

$$= \frac{r}{p} \sum_{n=r+1}^{\infty} \mathbb{P}(Y=n), \text{ where } Y \sim NB(r+1, p)$$

$$= \frac{r}{p}$$

Another POV: If $X \sim NB(r,p)$, then we can write $X = X_1 + X_2 + \dots + X_r$, where $X_i$ = # trials to go from $(i-1)$ succ to $i$ succ, each $X_i \sim Geom(p)$ : indep
$\Rightarrow$ By the linearity of expectation: $\mathbb{E}[X] = \sum_{i=1}^{r} \mathbb{E}[X_i] = \sum_{i=1}^{r} \frac{1}{p} = \frac{r}{p}$

Variance (Proof in HW)

$Var(X) = \frac{r(1-p)}{p^2}$

# HYPERGEOMETRIC DISTRIBUTION

Denoted as $X \sim Hypergeometric(N, m, n)$ (More related to binomial dist rather than geo dist)

$N, m, n \in \mathbb{Z}$, $0 \leq m \leq N$, $1 \leq n \leq N$

Setting

Population of size $N$, where $m$ are good
We select a random subset of $n$ members of the population w/o replacement uniformly
Count # good people

Values: $\{0, 1, \dots, \min\{m, n\}\}$

## OBSERVATION

If we sampled with replacement, then we have a bin dist $X' \sim Bin(n, \frac{m}{N}) \Rightarrow \mathbb{E}[X'] = \frac{nm}{N}$, $Var(X') = \frac{nm(N-m)}{N^2}$

# HYPERGEOMETRIC DISTRIBUTION QUANTITIES

Distribution

$$\mathbb{P}(X=k) = \frac{\text{\# ways of getting } k \text{ good} + n-k \text{ bad}}{\text{\# ways of getting } n} = \frac{\binom{m}{k}\binom{N-m}{n-k}}{\binom{N}{n}}$$

unif prob

Expectation

Let $X_i = \begin{cases} 1, & i^{th} \text{ in sample is good} \\ 0, & \text{o/w} \end{cases}$

Then, $X = \sum_{i=1}^{n} X_i$

By linearity of expectation, $\mathbb{E}[X] = \sum_{i=1}^{n} \mathbb{E}[X_i]$

Notice, $\mathbb{E}[X_i] = \mathbb{P}(i^{th}\text{ person is good}) = \frac{m}{N}$ because everyone is equally likely to be the $i^{th}$ chosen person (w/o prev knowledge)

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$\therefore \mathbb{E}[X] = \frac{nm}{N}$

## Variance

$\mathrm{Var}(X) = \mathbb{E}[X^2] - \mathbb{E}[X]^2$

$\hookrightarrow \mathbb{E}[X^2] = \mathbb{E}[(\sum_{i=1}^{n} X_i)^2] = \mathbb{E}[\sum_{i,j} X_i X_j] \overset{\text{linearity of expectation}}{=} \sum_{i,j} \mathbb{E}[X_i X_j] = \sum_{i=1}^{n} \mathbb{E}[X_i^2] + 2\sum_{i<j} \mathbb{E}[X_i X_j] = \sum_{i=1}^{n} \mathbb{E}[X_i] + 2\sum_{i<j} \mathbb{P}(i^{th}\text{ and }j^{th}\text{ person are good})$

$\because$ is 0 or 1

$= \frac{nm}{N} + 2\left[\frac{\#\text{pairs of good only}}{\#\text{pairs}}\right]\underbrace{\binom{n}{2}}_{i<j\text{ choices}}$

$= \frac{nm}{N} + 2\left[\frac{\binom{m}{2}}{\binom{N}{2}}\right]\binom{n}{2}$

$= \frac{nm}{N} + \frac{n(n-1)m(m-1)}{N(N-1)}$

$\therefore \mathrm{Var}(X) = \frac{nm}{N} + \frac{n(n-1)m(m-1)}{N(N-1)} - \frac{n^2m^2}{N^2} = \frac{nm(N-m)(N-n)}{N^2(N-1)}$ (Bin $\Rightarrow \frac{nm(N-m)}{N^2}$)

# GEOMETRIC DISTRIBUTION

Parameter: Success probability $0<p\le 1$, $X \sim \mathrm{Geom}(p)$

## Situation

Independent trials w/ success probability $p$

Count # trials until first success (inclusive)

Values: $k = 1, 2, 3, 4, \ldots$

# GEOMETRIC DISTRIBUTION QUANTITIES

$\mathbb{P}(X=k) = \underbrace{(1-p)^{k-1}}_{\text{first }k-1\text{ fails}}\underbrace{p}_{\text{last success}} \ge 0$

Notice, $\sum_{k=1}^{\infty} \mathbb{P}(X=k) = \sum_{k=1}^{\infty} p(1-p)^{k-1} = p \cdot \frac{1}{1-(1-p)} = 1$, so this is a valid probability ✓

Expectation: $\mathbb{E}[X] = \sum_{k=1}^{\infty} k\,\mathbb{P}(X=k) = \sum_{k=1}^{\infty} k(1-p)^{k-1}p = p\sum_{k=1}^{\infty} k(1-p)^{k-1}$

As $\sum_{k=1}^{\infty} x^k = \frac{x}{1-x}$, diff w.r.t. $x$, $\sum_{k=1}^{\infty} kx^{k-1} = \frac{(1-x)-x(-1)}{(1-x)^2} = \frac{1}{(1-x)^2}$ (☆)

$\therefore \mathbb{E}[X] = p\left(\frac{1}{(1-(1-p))^2}\right) = \frac{1}{p}$

Variance: $\mathrm{Var}(X) = \mathbb{E}[X^2] - \mathbb{E}[X]^2 = \mathbb{E}[X(X+1)] - \mathbb{E}[X] - \mathbb{E}[X]^2$

Diff (☆) w.r.t. $x$, $\sum_{k=1}^{\infty} k(k+1)x^{k-1} = -2(1-x)^{-3}(-1) = \frac{2}{(1-x)^3}$

$\therefore \mathrm{Var}(X) = \sum_{k=1}^{\infty} k(k+1)\mathbb{P}(X=k) - \frac{1}{p} - \frac{1}{p^2} = p\sum_{k=1}^{\infty} k(k+1)(1-p)^{k-1} - \frac{1}{p} - \frac{1}{p^2} = p\left(\frac{2}{p^3}\right) - \frac{1}{p} - \frac{1}{p^2} = \frac{1}{p^2} - \frac{1}{p}$

# EXAMPLE (COUPON COLLECTOR)

There are $n$ types of coupons.

Every coupon we get is unif random, indep of previous coupons.

How many coupons do we need to collect them all?

## Solution

Define $X_i$ := # purchases before we get coupon of type $i$, $1 \le i \le n$

Then, $X_i \sim \mathrm{Geom}(\frac{1}{n}) \Rightarrow \mathbb{P}(X_i = k) = \frac{1}{n}(1-\frac{1}{n})^{k-1}$, $\mathbb{E}[X_i] = n$

Now, let $X$ = # purchases until we have a full set

$\Rightarrow X = \max\{X_i \mid 1 \le i \le n\}$

Let $Y_i$ := # purchases to go from $(i-1)$ coupons to $i$ coupons

Notice, $X = \sum_{i=1}^{n} Y_i$, where $Y_1 \sim \mathrm{Geom}(1)$, $Y_i \sim \mathrm{Geom}(\frac{n-i+1}{n})$

success = seeing one of $n-(i-1)$ types of coupons we didn't see $\Rightarrow p = \frac{n-i+1}{n}$ (coupons: unif)

By linearity of expectation, $\mathbb{E}[X] = \sum_{i=1}^{n} \mathbb{E}[Y_i] = \sum_{i=1}^{n} \frac{n}{n-i+1} = n\sum_{i=1}^{n} \frac{1}{i} = nH_n$